AllMinds

Functional Components & Win-Readiness

How each feature makes its decisions, what works today, and the fixes that make AllMinds hackathon-winning

Neurodivergent-affirming AI teaching companion — HackAstone Agentic AI

Generated from the source of the HackAstone-Agentic-AI/AllMinds repository

1. Summary

I read the code behind all 17 user-facing functions of AllMinds and traced how each one reaches its result. **7 work as intended, 7 work partially, 2 do not work with the data the app itself captures, and 1 exists only in the API.** The idea, the architecture and the AI agents are strong; what would cost points in front of judges is a small number of concrete defects at the seams between screens, APIs and agents. **13 items are rated P0** (fix before the final). Most are an hour or two each.

How this was produced. Static reading of the frontend, backend and agents source. Nothing was run against live services, so statements describe what the code does when executed as written. Status labels: Working = does what it claims; Partial = works with a visible gap; Not working = cannot produce a correct result from data the app itself collects; No UI = backend only.

The five things judges are most likely to hit

- **Analytics shows zeros** after adding a progress record (W1), and the AI says every learner needs support (W8).
- **Support request does nothing** yet reports success (W2).
- **The NGO finder** returns three placeholder-looking organisations and only after granting location (W4).
- **A natural forum post is rejected** because it contains the word 'student' (W3).
- **An empty new account** and long AI waits make the first impression weak (W5, W9).

2. How the app makes decisions

Every feature decides in one of five ways. Knowing which one lets you explain the product honestly and tells you where it can fail. **Fixed rules and formulas** are predictable and explainable (support-priority score, task ordering). **Regex and keyword checks** are fast but brittle (guardrails, forum filter, calendar phrases, RAG trigger). **LLMs** (Groq routing, Gemini writing) handle language but need guardrails. **Database queries** decide what a user may see (everything filtered by teacher id). **External APIs** add reach and failure modes (YouTube, Google Calendar, image generation).

Who makes the decision in each feature, and does it work today?

Feature	Fixed rule / formula	Regex / keyword	LLM	Database query	External API	Status
Login / sign-up	●			●		Working
OAuth sign-in	●			●	●	Working
Dashboard (history, tasks)	●			●		Working
Learner upload (CSV/XLSX)	●			●		Partial
Progress + sensory logs	●			●		Partial
Learner analytics	●					Not working
Support request			●			Not working
Assessment planner		●	●		●	Working
Visual support		●	●		●	Working
Parent letters		●	●	●		Working
Schedule tracking	●			●	●	Partial
Thandi chat		●	●		●	Working
Calendar via chat		●	●		●	Partial
Community forum		●		●		Partial
Strategy library / NGO reviews				●		No UI
Training resources						Partial
NGO finder	●			●		Partial
Settings (profile, theme, language)	●			●		Working

Figure 1. Decision mechanism and current status for every feature (from code reading).

2.1 The decisions, one by one

Decision	How it is made	Where
Which agent answers	tool_hint from the page; otherwise Groq Llama-3.3-70B returns one word (temperature 0, 10 tokens) from a prompt with ~25 examples; regex extracts it; defaults to 'general'. No Groq key: keyword table, first match wins.	core/router.py, supervisor.py
Use curriculum context?	Lesson requests only, if the text contains grade, phase, curriculum, dcaps, caps, foundation, numeracy, literacy, life skills, home language or fal. Then the RAG agent retrieves 3 chunks of DCAPS.pdf.	router.py _needs_rag_ enrichment
Block a message?	Regex checks: length > 2000, prompt-injection phrases, self-harm/abuse/weapons phrases, off-topic abuse. Otherwise PII (13-digit IDs, e-mails, SA phones) is replaced with [REDACTED].	core/guardrails.py
Which material type	Page sends material_type (lesson_plan, worksheet, test, homework); a dict picks the system prompt.	lesson_planer_agent.p y
Progress or concern letter	Teacher's choice; if missing, 'concern' in the text means concern, otherwise progress.	parent_comm_agent.py _resolve_type
Learner risk band	Mean of three gaps: 100 - engagement%, 100 - task-completion%, 100 - attendance%. >= 70 High, >= 40 Moderate, else Low. Only available dimensions are averaged.	learnerAnalytics getProgressAnalytics
Trend and risk flags (agent)	Last value minus first: > +5 improving, < -5 declining. Flags: average mark < 50, average behaviour < 50, declining overall. Progress score = 0.7 x mark + 0.3 x behaviour.	analytics_automation.p y
Is a visual output valid	Must contain 4 headings; otherwise the model is called once more.	visual_support_agent.p y
Which image model	IMAGE_MODEL first, then a cached winner, then 3 fallbacks; stops on billing or quota errors.	visual_media.py
Nearest NGOs	Haversine distance to every NGO with coordinates, sort ascending, top 6, optional service substring filter.	professional-developm ent page
Task status	pending -> in_progress -> completed on each click; open tasks sorted by due date, undated last.	dashboard page
Forum post allowed?	Rejected if title+text contains name, address, phone, email or student (substring, lowercase).	forum page
Calendar action	Phrase lists decide list vs create; first number in the text is the time; 'tomorrow'/'today' set the day.	google_calendar_auto mation.py
Learner display name	First non-empty of pseudonym, preferred_name, learner_name, student_name, display_name, full_name, name; else first + surname; else anonymous id.	lib/learners.ts

Table 1. Decision rules extracted from the code

3. Feature catalogue

Feature	What it does	How it decides	Status	Main gap
Landing, login, sign-up, OAuth	Public pages; e-mail/password or Google/Microsoft/Apple.	Fixed rules: password length, bcrypt compare, one-time OAuth code.	Working	Single-word OAuth names can fail (profile NOT NULL).
Dashboard	Six cards (learners, plans, letters, activity, tasks, scheduled work), search, task board.	Counts and filters computed in the browser; tasks sorted by due date; status advances pending -> in progress -> completed.	Working	Load errors are swallowed and shown as 'nothing yet'.
Learner profiles (upload)	Import CSV/XLSX class lists (max 5 MB, 1000 rows), list, export.	Header normalisation; ID columns recognised by name; everything else becomes JSON profile_data.	Partial	No PII warning; export has IDs and dates only.
Progress and sensory logs	Per-learner records of learning area, level, achievements, behaviour, triggers.	Teacher input; 'AI suggestions' are two fixed template sentences.	Partial	Engagement, attendance and task completion are never collected.
Learner analytics	KPIs, bars and AI narrative per learner.	Support priority = mean(100 - engagement%, 100 - completion%, 100 - attendance%); High >= 70, Moderate >= 40.	Not working	Inputs are empty, so every learner shows zeros / Low.
Support request	Send a goal + context to the AI for a learner.	Intended: LLM; actual: analytics agent, which ignores the text.	Not working	Context dropped, reply discarded.
Assessment planner	Lesson plan / worksheet / test / homework with videos and images; save and PDF.	tool_hint 'lesson'; keyword test triggers DCAPS retrieval; Gemini 2.5 Pro writes; YouTube + image threads.	Working	25-90 s with only a spinner.
Visual support	Visual supports, interactive idea, quick-make tip.	tool_hint 'visual'; output validated for 4 headings, one retry.	Working	Same wait as the planner.
Parent communication	AI drafts progress or concern letters, edit, save, export.	Type chosen by the teacher; latest progress and sensory entry added to the prompt.	Working	.doc export does not escape HTML.
Schedule tracking	Per-learner daily timeline; optional Google Calendar copy.	Time must match HH:MM; unique per learner and time; calendar copy created for today.	Partial	A list, not a visual schedule; no day or recurrence.
Thandi chat (floating)	Ask anything, in any SA language.	No hint, so Groq Llama-3.3-70B picks one of 8 agents; keyword fallback.	Working	No memory, no learner context, no streaming.
Calendar via chat	'Schedule...' / 'what's on my calendar'.	Phrase lists, then regex time parsing.	Partial	First number becomes the time; no confirmation.
Community forum	Posts and comments.	Browser blocks posts containing name, address, phone, email or student.	Partial	Blocks ordinary posts (substring match).
Strategy library and NGO reviews	Teacher-shared strategies, votes, ratings.	Backend and tables exist.	No UI	Nothing in the frontend calls them.
Training resources	Six course cards.	Hard-coded list.	Partial	Links go to provider home pages.
NGO finder	Nearest support organisations.	Haversine distance, nearest 6, optional service text filter.	Partial	3 placeholder-looking records; location-only.
Settings	Profile, language, theme, high contrast, Google Calendar connect.	Stored in users/teacher_profiles and localStorage.	Working	UI language list is complete but only 56 UI strings exist, and most are untranslated per language.

Table 2. All functional components

4. Deep dives

4.1 NGO finder

This is the feature most likely to be judged on real-world impact, so it needs to feel trustworthy. Today it is a client-side nearest-six lookup. The organisations table has the right fields (services, province, city, coordinates, rating, verified) and the API already supports province, city, service and text search, but the page does not use them.

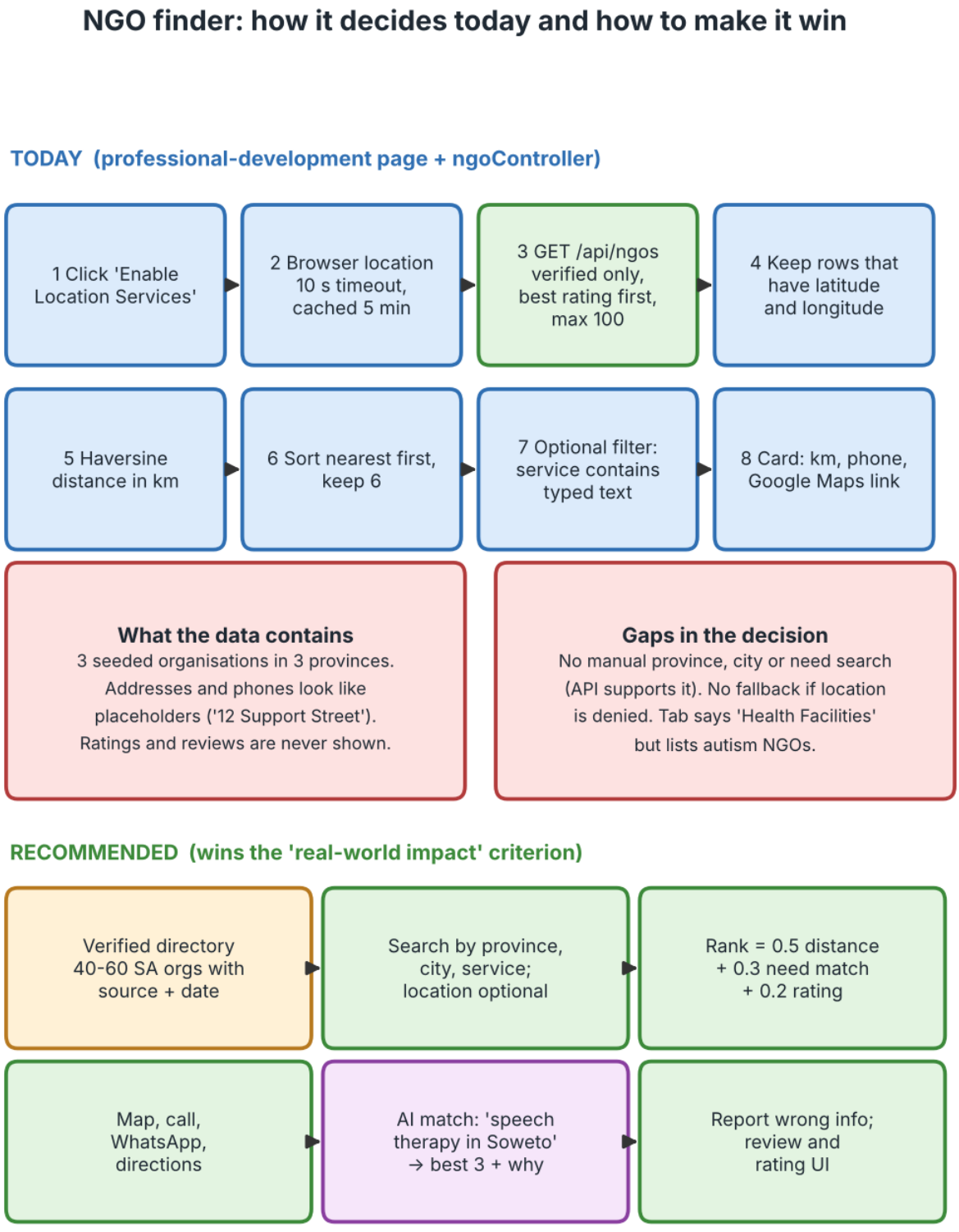


Figure 2. The finder's decision chain and the recommended version.

Recommended ranking (simple, explainable, easy to defend): score = 0.5 x (1 - distance / 100 km, floored at 0) + 0.3 x (service matches the learner's need) + 0.2 x (rating / 5). Show why a result ranked high ('12 km away - speech therapy - 4.5 stars'). Verify every record against the organisation's own website or by a call, store the source URL and last-checked date, and show it. Candidate national bodies to research (confirm each before listing): Autism South Africa, ADHD South Africa, the Dyslexia Association of South Africa, and the South African Federation for Mental Health, plus provincial branches and public clinics.

4.2 Learner analytics

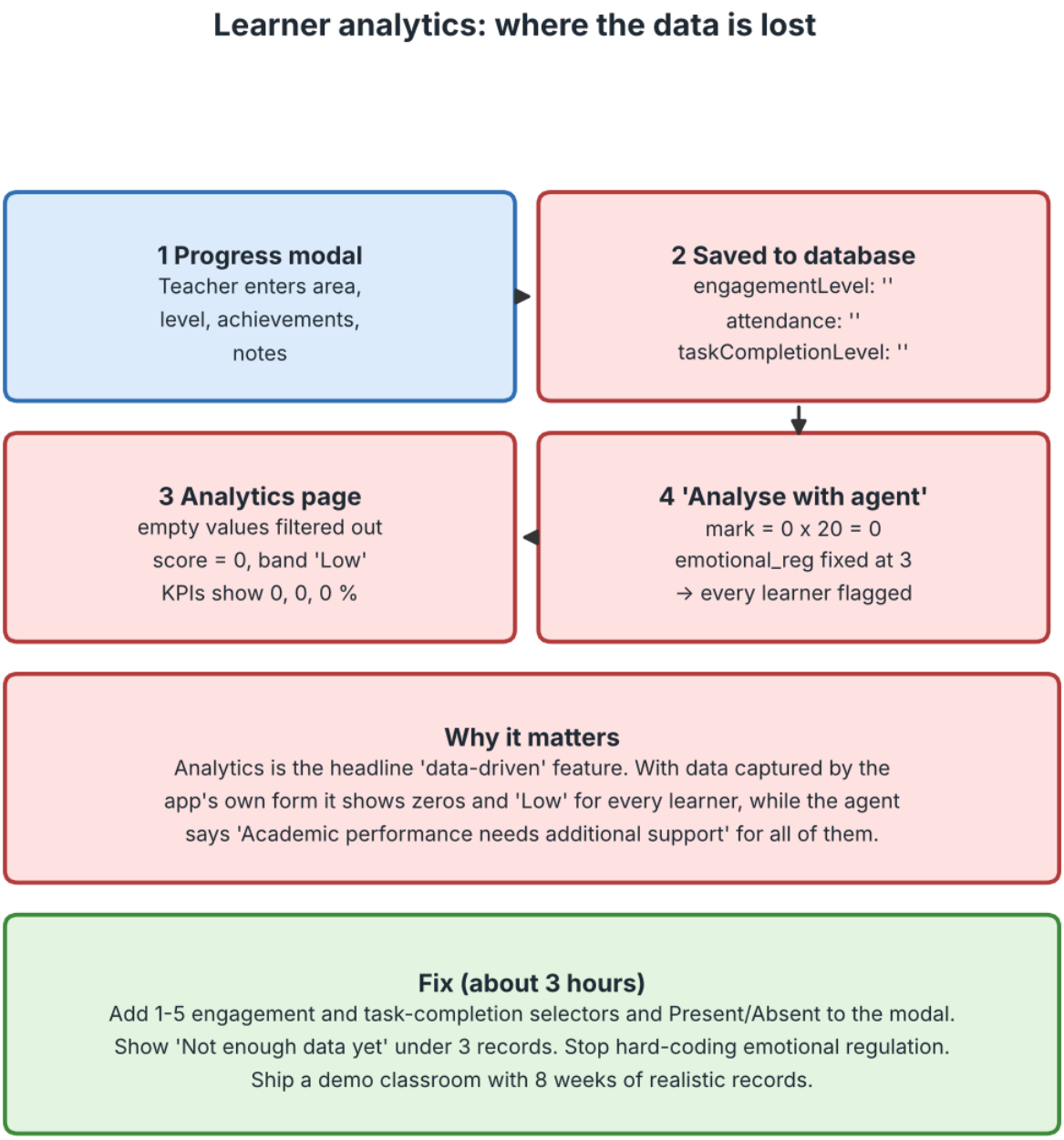


Figure 3. The analytics inputs are never collected by the UI, so the output is always zero.

The scoring itself is sound and explainable. The problem is purely data capture and a few invented constants. Once W1 and W8 are fixed this becomes one of the strongest demo screens: a ranked class view, trends, and a plain-language narrative.

4.3 Support request



Figure 4. Every hop forwards the request but nothing uses it.

4.4 Calendar and schedule

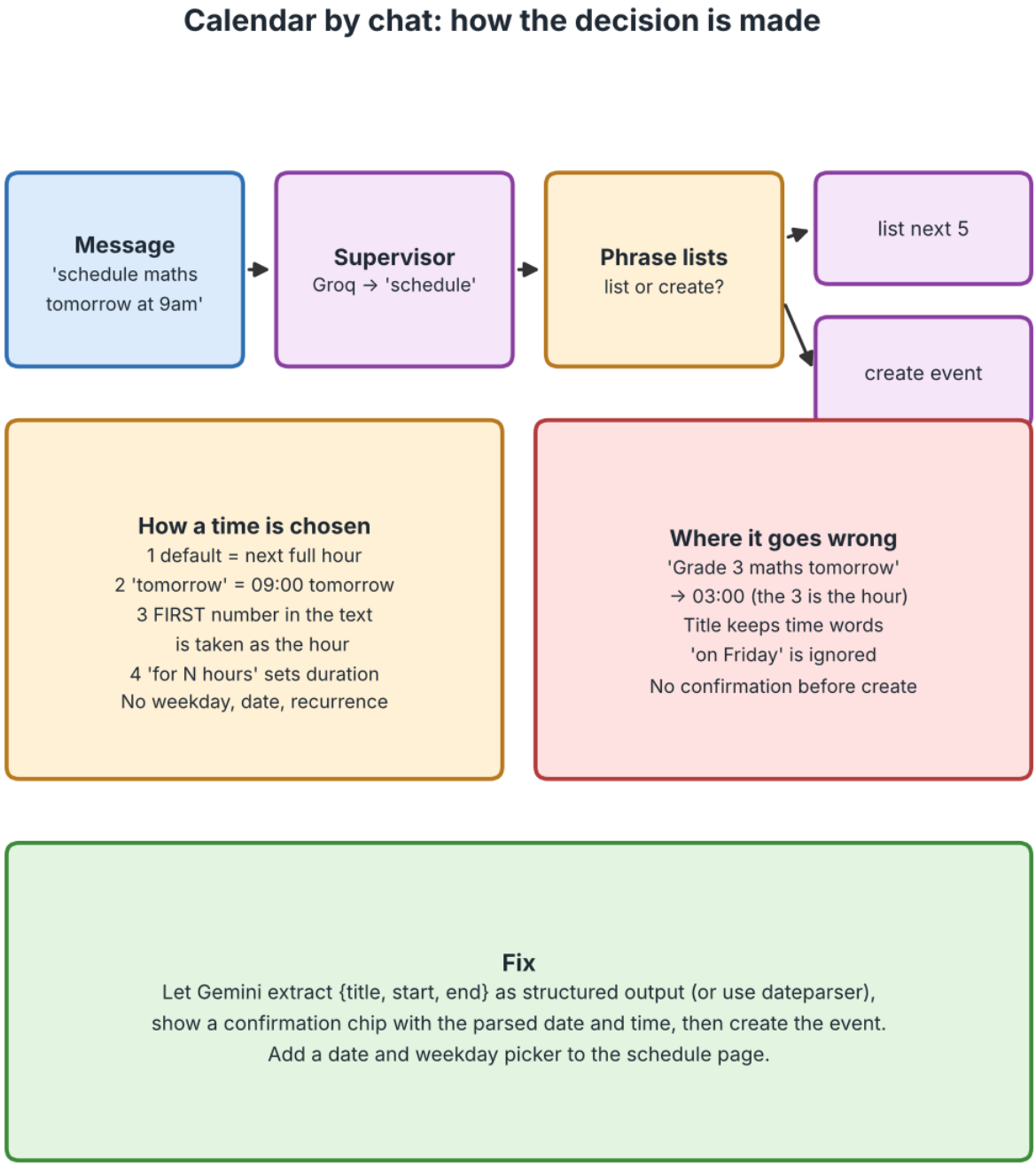


Figure 5. Decision flow for 'schedule ...' in the chat, and its failure modes.

4.5 Other decisions worth knowing

- **Forum PII filter:** the intent is right (protect children) but the implementation blocks the word 'student'. A regex for IDs, phone numbers, e-mails and capitalised name pairs, reused from guardrails.py and enforced in the backend, fixes it.
- **Parent letters:** the page builds the prompt from the latest progress and sensory entries, which is good. Because all context travels as text in the message, long data hits the 2000-character guardrail limit; pass learner context as structured fields instead.
- **Planner:** the curriculum is chosen on the page but 'curriculum' is dropped by ChatRequest; it only reaches the model because it is also written into the prompt text.
- **Dashboard:** everything is computed in the browser from five parallel calls; good for speed, but it needs error states and could show an AI 'today's priorities' card (X3).

5. Issues to fix so AllMinds can win

P0 = fix before the final (breaks or undermines a headline feature). P1 = credibility and polish. P2 = optional features that differentiate you. Items R1-R4 come from the Production Readiness review because they can fail live. Effort is a rough estimate for one developer.

5.1 P0: fix before the final

ID	Pri	Area	Issue	Suggested fix	Effort
W1	P0	Analytics	Progress form never captures engagement, attendance or task completion (hard-coded empty strings), so analytics shows zeros and 'Low', and the AI agent flags every learner as struggling.	Add 1-5 selectors and Present/Absent; 'not enough data' state under 3 records; map sensory logs to emotional regulation.	3 h
W2	P0	Analytics	'Create support request' cannot work: unknown fields are dropped by ChatRequest, the analytics agent ignores the message, and the response is thrown away while the UI says to review it elsewhere.	Add fields to ChatRequest, route by request_type, put context in the prompt, show the reply inline.	4 h
W3	P0	Forum	Post filter rejects any text containing 'name', 'student', 'phone', 'email' or 'address' (substring), including 'students', 'username', 'renamed'. Most real posts fail.	Detect real PII (SA ID, phone, e-mail, capitalised full names) with regexes, share the guardrail with the agents, enforce on the server.	1 h
W4	P0	NGO finder	Only 3 seeded organisations with placeholder-looking contact details, found only through browser location; no manual search, no fallback if location is denied, ratings and reviews hidden.	Curate 40-60 verified SA organisations with source and date, add province/city/service search, map, call/WhatsApp, rating UI, and the ranking described in section 4.1.	1 day
W5	P0	Demo	A new account is empty on every page, and the demo user's password hash is committed in the baseline schema (also seeded into any production database).	'Load demo classroom' button that creates 10 learners with 8 weeks of records, plans and letters; remove the seeded user outside development.	4 h
W6	P0	UX	Failed API calls are only logged to the console, so pages show friendly empty states ('No learner profiles yet') instead of an error.	Error banner with Retry, skeleton loaders, shared useApi hook.	2 h
W7	P0	Calendar	Chat calendar parser reads the first number as the time ('Grade 3' becomes 03:00), keeps time words in the title, has no weekday or date support and creates events without confirmation. The schedule page copies items to today only.	Gemini structured extraction or dateparser, confirm chip before create, date and weekday fields on the schedule page.	4 h
W8	P0	Analytics	Agent computes narrative, risk flags and chart data but returns only a one-line summary; mark is engagement x 20 and emotional regulation is hard-coded to 3.	Return structured JSON, remove invented constants, render narrative and flags in the UI.	3 h
W9	P0	Latency	Planner and visual tool take 25-90 s with a plain spinner; two judges at once can stall the agent service (blocking handler).	Staged progress text, cached sample outputs for the demo topics, non-blocking endpoint (see R3).	4 h
R1	P0	Release	Migration 20260629 uses ADD CONSTRAINT IF NOT EXISTS (invalid in PostgreSQL) and blocks the deploy step.	DO $$ block checking pg_constraint.	30 min
R2	P0	Sign-in	teacher_profiles name/surname NOT NULL vs NULL inserts (single-word names).	Make nullable or default ''.	30 min
R3	P0	Agents	Blocking LLM calls inside the async handler: one 60 s lesson stalls health checks and other users.	def endpoint / threadpool, 2+ workers.	1 h
R4	P0	Agents	RAG index rebuilt at first use: 30-120 s first answer and duplicate vectors.	Prebuild the index, load on startup, warm in /health.	3 h

Table 3. P0 issues

5.2 P1: credibility and polish

ID	Pri	Area	Issue	Suggested fix	Effort
W10	P1	Honesty	'AI suggestions' in the progress modal are two fixed template sentences stored as aiSuggestions.	Call the general agent with the record, or relabel as 'Tips'.	1 h
W11	P1	Training	Courses are a hard-coded list linking to provider home pages; titles may not match real courses.	Link to verified course pages and South African DBE/inclusive-education resources; store in the database.	2 h
W12	P1	Community	Strategy library with votes and NGO reviews exist in the API and database but have no screens.	Build a simple strategy library (filter by challenge, vote) and NGO review form, or remove from the pitch.	1 day
W13	P1	Privacy	Imported CSV columns are stored as-is (could hold real names or ID numbers) although the UI says learners are anonymous; export contains IDs and dates only and is not CSV-escaped.	Import preview with PII detection and column mapping, template download, proper export.	4 h
W14	P1	i18n	Language picker lists 11 languages but the UI has 56 strings and the non-English tables are partial. AI replies are multilingual; the interface mostly is not.	Say 'AI answers in 11 languages' in the pitch, translate the top 60 strings, fall back gracefully.	1 day
W15	P1	Security	Downloaded .doc letters inject letter text into HTML without escaping.	Escape HTML or build a real docx.	30 min
W16	P1	Schedule	Schedule is a time list. Neurodivergent learners need a visual, now/next timeline they can see and print.	Timeline cards with icons, now/next banner, print view, 'generate visual schedule' via the visual agent.	5 h
W17	P1	Thandi	No memory, no learner context, no streaming, conversation lost on reload.	Wire the SQLite/LangGraph memory (see dependencies), pass selected learner, stream tokens.	1 day

Table 4. P1 issues

5.3 P2: wow factor

ID	Pri	Area	Issue	Suggested fix	Effort
X1	P2	Wow	Impact dashboard for judges: plans generated, hours saved, learners supported, languages used.	Count rows already in activities/lesson_plans; add one card row.	3 h
X2	P2	Wow	AI NGO matcher: 'my learner needs speech therapy in Soweto' returns the 3 best organisations with reasons.	RAG over the directory with the agents' existing pattern.	5 h
X3	P2	Wow	'Who needs attention today' ranked list from the support-priority score and recent logs.	Reuse getProgressAnalytics across learners once W1 is fixed.	3 h
X4	P2	Wow	One-click class pack: lesson plan + visual support + parent letter bundled as one PDF.	jsPDF already present; call three agents in parallel.	5 h
X5	P2	Wow	Voice in / read aloud for Thandi (Web Speech API) and streaming replies.	Browser APIs; SSE from FastAPI.	5 h
X6	P2	Wow	Source citations (DCAPS page numbers) on curriculum-grounded answers.	Return chunk metadata page numbers and render them as chips.	2 h

Table 5. P2 ideas



Figure 6. A one-week plan that fits the time left.

6. Five-minute demo script

Time	Show	Say
0:00	Landing page, sign in with the demo classroom loaded (W5).	Teachers have neurodivergent learners and no specialist support.
0:45	Dashboard: learners, recent plans, tasks.	Everything is scoped to the teacher; learners are pseudonymised.
1:15	Assessment planner: Grade 3 maths, fractions, dyscalculia, isiZulu. Show a cached result, then the live progress steps.	Supervisor-worker agents, Gemini Pro, DCAPS retrieval, parallel videos and images.
2:15	Learner analytics: ranked class view, one learner's trend, narrative.	Transparent formula, not a black box.
3:00	Parent letter drafted in the learner's home language, export PDF.	Hours of admin saved per week.
3:45	NGO finder: search by need and province; AI match.	Closing the loop from classroom to community support.
4:30	Thandi answers a curriculum question (spoken input and citations if X5/X6 are built).	Safe guardrails, AI replies in 11 languages.

Table 6. Demo script

7. Judging questions and model answers

These answers assume the P0 and listed P1/P2 fixes are done. Do not claim a capability in the room until it is built and tested.

Q1. How does the NGO finder decide what to show?

Today it takes the teacher's location, fetches verified organisations, computes great-circle distance, and shows the nearest six with an optional service filter. The next version ranks by distance, match to the learner's need and rating, shows why, and works without location through province and city search.

Q2. How does the support-priority score work, and can you trust it?

It averages how far engagement, task completion and attendance fall short of 100%, and bands it at 40 and 70. It is a screening aid built from teacher-entered ratings, not a diagnosis, and the UI should state that next to the score. It is explainable: every point of the score can be traced to a logged value.

Q3. Where is AI actually deciding, and where is it deliberately not?

AI decides routing (Groq), writes lesson content, letters and visual supports (Gemini) and answers curriculum questions from retrieved text. Risk scoring, trends and task ordering are deterministic rules on purpose, so teachers can audit them.

Q4. What stops the AI giving harmful advice?

Input guardrails (injection, harm, abuse, PII), prompts that forbid diagnosis and medication advice, output filtering, and signposting to specialists. For safeguarding disclosures (to be added) it routes to a vetted script instead of a refusal.

Q5. Is the data real?

The demo classroom is synthetic and labelled so. The NGO directory is curated from verified public sources with the source and last-checked date shown on each card.

Q6. What did you learn from reviewing your own product?

That the risky parts are the seams: forms that do not collect what analytics needs, payload fields that drop between services. We are closing them with end-to-end tests that follow one learner from data entry to insight.

Q7. What would you add with another month?

Memory for Thandi, offline mode for low-connectivity schools, a school-level dashboard for principals, and an evaluation set that measures quality per agent.